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SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING

Academic Year: 2023-2024

Program: MBA Tech. Stream: Data Science Year: II Semester: IV

Subject: Stochastic Processes and Applications Time: 3 hrs (10.00 a.m to1.00 p.m)

Date: 25 / 6 / 2024 No. of Pages: 03

Marks: 100

Re-Examination 2022-23

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. 1 is compulsory.
2) Out of remaining questions, attempt any 4 questions.
3) **In all 5 questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

Q1		Answer briefly: (5 marks each)	[20]
CO-1; SO-1; BL-2	a.	Brief on autoregressive process and its application in the analysis of random processes. The explanation shall refer to the mathematical expression of autoregressive process.	
CO-2; SO-1; BL-4	b.	The lifetime of certain brand of electric bulb may be considered a random variable with mean 1200 hours and standard deviation 250 hours. Using Central Limit theorem estimate the probability that the average lifetime of 60 bulbs exceeds 1250 hours.(Z value area between 0 and 1.55 = 0.4394)	
CO-1; SO-1; BL-3	c.	In box 1 there are 3 white and 7 black balls, in box 2 there are 4 white and 6 black balls. A box is selected at random and a ball is drawn. Evaluate the probability that the ball will be white	
CO-3; SO-1; BL-3	d.	A random process is given by $X(t) = A\cos(\omega_0 t + \theta)$ where θ is a random variable uniformly distributed over $(0, \pi)$. Find the expectation of X(t).	
		Answer any four from Q2 to Q7	

Q2 CO-4; SO-1; BL-3	a	If X denotes the outcome when a fair die is tossed, compute m.g.f of X and hence, find the mean and variance of X.	[10]
CO-3; SO-1; BL-2	b	Compare Random Process and Random Variable. Explain the Classification of Random Processes	[10]
Q3 CO-4; SO-1; BL-3	a	A continuous function of a random variable X has the p.d.f $f(x) = kx^2e^{-x}$, x>=0. Calculate k, mean and variance.	[10]
CO-2; SO-1; BL-2	b	Explain in detail Deterministic and Non Deterministic, Ergodic Process, Markov Process of Random (Stochastic) Processes. Explain first order statistics	[10]
Q4 CO-3; SO-1; BL-3	a	A random process is given by $X(t) = A\cos(\omega t) + B\sin(\omega t)$ where A and B are random variables. Show that $X(t)$ is wide sense stationary if the expectations are given (i) $E(A) = E(B) = 0$ (ii) $E(A^2) = E(B^2)$ and (iii) $E(AB) = 0$.	[10]
CO-4; SO-1; BL-3	b	Suppose, a drunkard (man) wants to travel to his home by walking a 4 feet wide road. Let X_n be the distance measured (in feet) from the n^{th} position on the road to the left end of the road. The man, as he is drunk, can move 1 foot to the left with probability 'p' (0<p<1) and right with probability 'q' (q=1-p). The man will fall down if he goes to the left end (X_n=0) or right end (X_n=4) of the road. The distance to his home is 7 feet long. Mention the transition probability matrix (TPM) and show that the random walk of the man is a Markov chain. Also, find the probability that the drunkard moves to left by 2 feet from mid of the road.	[10]
Q5 CO-3; SO-1; BL-3	a	The transition probability matrix (TPM) of Markov chain $\{X_n\}_{n=1,2,3,..}$ with three states is as follows. $$\begin{array}{cc} & \text{States of } X_{n+1} \\ & \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ \text{States of } X_n \begin{array}{c} 1 \\ 2 \\ 3 \end{array} & \begin{bmatrix} 0.2 & 0.5 & 0.3 \\ 0.5 & 0.4 & 0.1 \\ 0.3 & 0.4 & 0.3 \end{bmatrix} \end{array}$$ The initial distribution P^0 is as follows. $$\begin{array}{cc} & \begin{array}{ccc} 1 & 2 & 3 \end{array} \\ P^0 = & \begin{bmatrix} 0.6 & 0.3 & 0.1 \end{bmatrix} \end{array}$$	[10]

		Find $P(X_2 = 3)$, $P(X_2 = 2)$, and $P(X_3 = 2,\ X_2 = 3,\ X_1 = 3,\ X_0 = 2)$	
CO-4; SO-1; BL-3	b	A random process is given by $X(t) = \cos(\omega t + \Theta)$ where Θ is uniformly distributed in the interval $(-\pi, \pi)$. Calculate the mean, autocorrelation function and autocovariance function.	[10]
Q6 CO-3; SO-1; BL-3	a	A random process is given by $X(t) = A\cos(t + \emptyset)$ where $\emptyset$ is a random variable uniformly distributed over $(0, 2\pi)$. Check whether $X(t)$ is mean-ergodic (on the first order) and find the autocorrelation function of $X(t)$.	[10]
CO-2; SO-1; BL-3	b	The average number of visits at an amusement park is 30 children in one hour. What is the probability that (i) there are no visits in a 3-minutes time span, (ii) at least 5 visits in a 5-minutes time span?	[10]
Q7 CO-3; SO-1; BL-3	a	Compare wide sense stationary (WSS) and strict sense stationary (SSS) processes. Also, if a random process $X(t)$ is wide sense stationary with the autocorrelation function $R_X(\tau) = 9e^{-\alpha\|\tau\|}$ where $\tau = t_1 - t_2$. Find the second order moment of the random variable $X(9) - X(3)$. What is the value of the computed second order moment at $\alpha = 5$?	[10]
CO-1; SO-1; BL-2	b	Write a note on continuous time Markov chain.	[10]